



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PURE MATHEMATICS
PAPER 2

6042/2
3 hours

JUNE 2026 SESSION

Additional materials:
Answer booklet
Graph paper
List of Formulae MF 7
Scientific calculator [Non-programmable]

8050

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer booklet.

Answer **all** questions in Section A and any **five** questions from Section B.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

If a numerical answer cannot be given exactly and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

This question paper consists of 7 printed pages.

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[Turn over



Section A [40 marks]

Answer all questions in this section.

1. Solve the inequality.

$$\frac{x-3}{x+7} > 0.$$

[5]

2. It is given that

$$f(x) = |3x + 2|.$$

- (a) Sketch the graph of

$$f(x) = |3x + 2|.$$

[2]

- (b) Hence or otherwise solve the inequality.

$$f(x) \leq 2$$

[3]

3. (a) Solve the equation

$$\frac{2^{x+2} \cdot 16^{x+1}}{64^{x+3}} = \frac{1}{4096}.$$

[4]

- (b) The volume of a cylindrical bee-hive, $V \text{ cm}^3$, varies directly as the square of the radius $r \text{ cm}$ and its height $h \text{ cm}$.

- (i) Find the percentage change in V , if its radius increases by 2% and the height decreases by 4%.

[4]

- (ii) State whether the percentage change in V is an increase or a decrease.

[1]

4. A circle passes through the points A (3; -1), B (5; 2).

It is given that the circle touches the line $y = -x$ at the point P (1; -5).

Find the equation of the circle.

[10]



5. (a) It is given that
 $g(x) = 2x^3 + 2x^2 - 13x + 12$.
The remainder when $g(x)$ is divided by $(x + p)$ is 3 times the remainder
when $g(x)$ is divided by $(x - p)$.

Show that

$$2p^3 + p^2 - 13p + 6 = 0. \quad [5]$$

- (b) Solve the equation

$$2p^3 + p^2 - 13p + 6 = 0. \quad [6]$$



Section B [80 marks]

Answer any five questions in this section. Each question carries 16 marks.

6. (a) Prove by induction that

$$\sum_{r=0}^n p^r = \frac{1 - p^{n+1}}{1 - p} \text{ for } p \neq 1 \text{ and } n \geq 0. \quad [7]$$

- (b) The functions $g(x)$ and $h(x)$ are defined by $\frac{x}{3} - 1$ and $7x + 1$, respectively for $x \in \mathbb{R}$

(i) Find $hg(x)$ and hence evaluate $hg(4)$, [3]

(ii) Verify that $(hg)^{-1} = g^{-1}h^{-1}$. [6]

7. Given the polynomial

$$P(x) = x^4 - 4x^3 + 2x^2 + 4x - 3,$$

(a) (i) show that $x^2 - 1$ is a factor of $P(x)$, [2]

(ii) factorise $P(x)$ completely. [2]

(b) (i) Hence or otherwise express $\frac{x^2 - 1}{x^4 - 4x^3 + 2x^2 + 4x - 3}$, in partial fractions. [4]

(ii) By using the method of difference or otherwise, evaluate

$$\sum_{r=4}^7 \frac{r^2 - 1}{r^4 - 4r^3 + 2r^2 + 4r - 3}. \quad [4]$$

(iii) Using the result from b(i) evaluate

$$\int_4^7 \frac{x^2 - 1}{x^4 - 4x^3 + 2x^2 + 4x - 3} dx$$

leaving your answer in the form $k \ln b$. [4]



8. (a) Sketch on a single Argand diagram the region satisfying both $|z - 1 - i| \leq 2$ and $0 \leq \arg z \leq \frac{\pi}{4}$. [3]
- (b) Find the centre and radius of a circle represented by the locus $|z - 2 - i| = 3|z + 6 + 3i|$. [4]
- (c) Find the complex number z , which satisfies the equation $2z + i\bar{z} = 4 - i$, giving your answer in the form $x + iy$. [4]
- (d) The equation $4f^4 - 8f^3 + 9f^2 - 2f + 2 = 0$, has a root $1 - i$.

Find the other roots. [5]

9. Three points $A(1; 2; 1)$, $B(2; -1; -4)$ and $C(1; 0; -1)$ lie on the plane ABC. Point D has coordinates $(3; 4; 5)$.

Plane π has equation $r \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = 4$.

Find the

- (a) vector equation of line $\overrightarrow{CD}$, [2]
- (b) shortest distance from origin to π , [2]
- (c) angle between the line $\overrightarrow{BD}$ and line $\overrightarrow{CD}$, [3]
- (d) point of intersection of line $\overrightarrow{CD}$ with plane π . [4]
- (e) vector equation of the plane ABC. [5]



10. (a) The transformation M , given by $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ mapped points $P(2;1)$, $Q(4;1)$ and $R(3;5)$ onto P' , Q' and R' , respectively.

- (i) Find the coordinates of the image points P' , Q' and R' . [3]
 (ii) Interpret the transformation represented by M . [2]

(b) Matrix $A = \begin{pmatrix} -1 & 2 & 3 \\ 2 & -1 & 4 \\ 1 & 1 & -1 \end{pmatrix}$.

- (i) Find A^{-1} . [8]
 (ii) Hence or otherwise solve the simultaneous equations.
 $-x + 2y + 3z = 1$
 $2x - y + 4z = 0$
 $x + y - z = 2$ [3]

11. (a) It is given that $\frac{dy}{dx} = \sqrt{ye^x}$ and $y = 0$ when $x = 0$.

Show that

$$y^{\frac{1}{2}} = e^{\frac{1}{2}x} - 1. \quad [5]$$

- (b) A cylindrical container of depth x_0 cm is initially full of water. The water leaks through the bottom of the container at a rate proportional to the square root of the depth of the water. The depth of the water at time t hours after the leak started is x cm.

- (i) Write down the differential equation connecting x and t . [2]
 (ii) If the container is exactly half full after one hour, find the time it takes to be completely empty. [9]



12. (a) Given that $y = x$ and $y = x^3$, for $x \in \mathbb{R}^+$,

(i) Sketch on the same axes, the graphs of $y = x$ and $y = x^3$ for $x \in \mathbb{R}^+$. [2]

(ii) The region R is bounded by the curve $y = x^3$ and the line $y = x$ from $x = 0$ to a point A, where the line and curve intersect. Find the coordinates of A. [1]

(iii) Find the area of the region R. [3]

(iv) Region R is rotated through π radians about the x - axis. Find the exact value of the volume of the solid formed in terms of π . [4]

(b) It is given that A and T are positive integers. The binary operation (a) is defined by $A (a) T = 2A + T - 4$.

(i) Find $4 (a) 5$ [1]

(ii) Determine whether or not the binary operation (a) in the set of positive integers, forms a group. [5]

